

DNA Structure: Pair, Twist, Critique

LAUNCH • Science • 40 minutes

I can construct a complementary DNA section and describe the strengths and limitations of the model.

Name: _____ Date: _____

You may write, type, say, sign, point or direct a partner. An adult can read the words and record your exact response. Keep the choice and explanation your own.

Words for this lesson

Word	Meaning
nucleotide	one DNA building unit
complementary	fits by a pairing rule
double helix	two-strand twist

First idea

One DNA strand reads A-C-G-T. Which complementary strand fits the pairing rule?

- A. T-G-C-A
- B. A-C-G-T
- C. A-A-A-A

My first idea and reason:



A T pairs

We do • choose, explain, check

A short strand reads G-A-T-C. Which complementary sequence is correct?

A. C-T-A-G

B. G-A-T-C

C. A-C-T-G

My choice: _____ Evidence or reason:

Complementary base-pair lab

Match each base to its complementary partner, freeze the sequence, then critique one feature of the model.

Read or hear one card. Choose a group. Give the evidence. A partner checks your reason before you move on. Point or write card numbers; cutting is optional.

Group	Card numbers and evidence
A ↔ T PAIR	
C ↔ G PAIR	
MODEL LIMIT	

Activity cards

1 • Adenine	2 • Cytosine
Base A needs its complementary partner.	Base C needs its complementary partner.
3 • Not to scale	

The classroom pieces are far larger and farther apart than atoms and molecules.

Read the two sequences, then name what the model helps us see and what it cannot show.

Your lesson record

Construct a complementary DNA section, annotate its structure and evaluate the model.

Model helps us see

The manipulable ladder makes complementary pairing and two-strand structure visible and checkable.

Model does not show

The model is enormously enlarged, uses arbitrary colours, simplifies chemical bonds and cannot show the full chromosome or molecular dynamics.

Supported

Complete a six-base sequence with A-T and C-G cards, then point to the two strands.

Pairing key stays visible; one base at a time and exact scribing are allowed.

Standard

Complete a ten-base complementary sequence and describe nucleotide, two strands and double helix.

Use: Base ___ pairs with ___; the two strands ___.

Stretch

Construct and check the sequence, then evaluate two useful features and two limitations of the model.

Do not add Higher-tier protein synthesis; deepen model evaluation.

Evidence target

A correct complementary sequence, labelled two-strand/double-helix description and one valid model limitation.

Exit, Audience and evidence • W12L2

Supported: Which base pairs with C?

Standard: Write the complement of A-G-C-T.

Stretch: State one strength and one limitation of today's DNA model.

What I said, and what it changed

Next I will _____.

Adult witness note

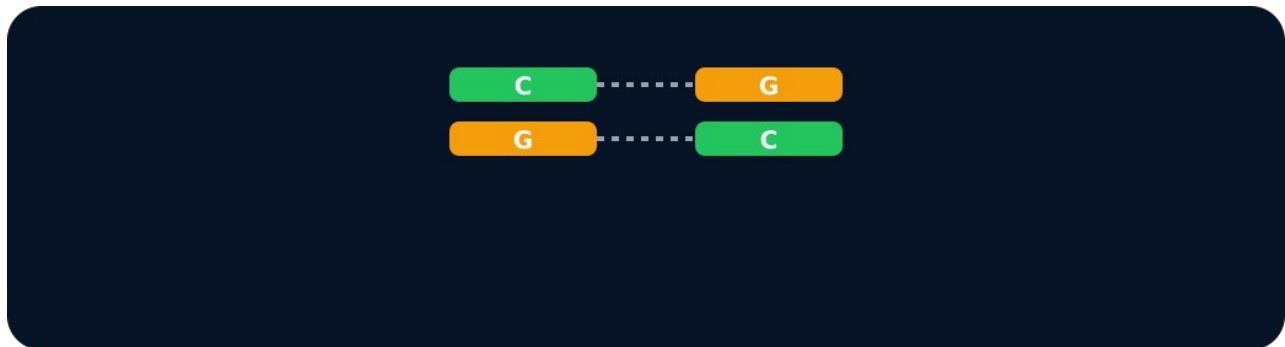
What was observed, and with what level of independence?

My evidence and explanation

What does this evidence not show?

Visual reference cards

These are diagrams or model views from the uploaded lesson. Use their labels and the card descriptions; they are not photographs of your own results.



C G pairs



Ladder

Exit • one next step

After the adult has listened, what will you keep, improve or check next?